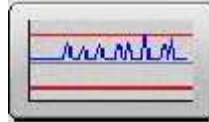


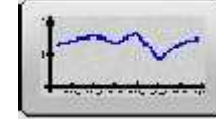
Home Screen/Production Pop-Up Menu



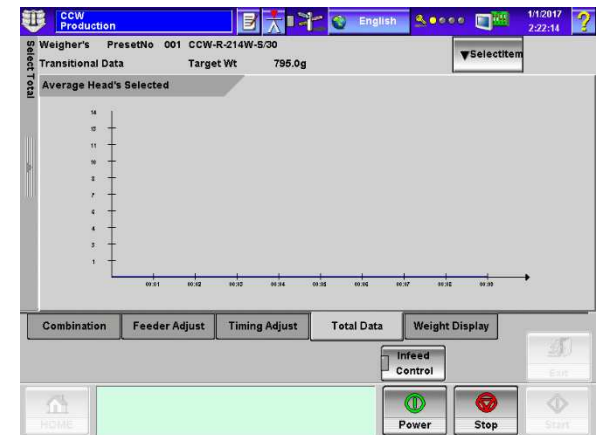
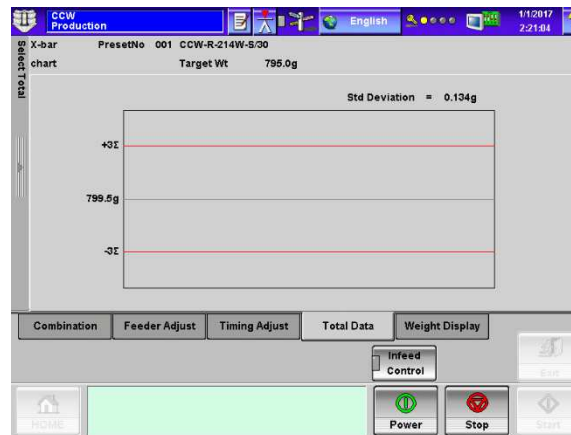
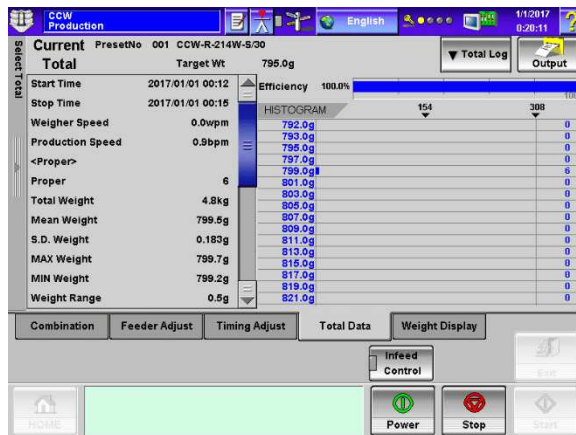
Histogram



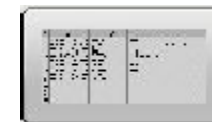
X-bar Chart



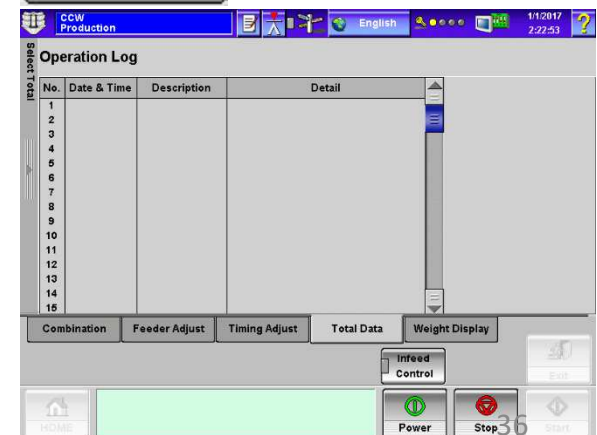
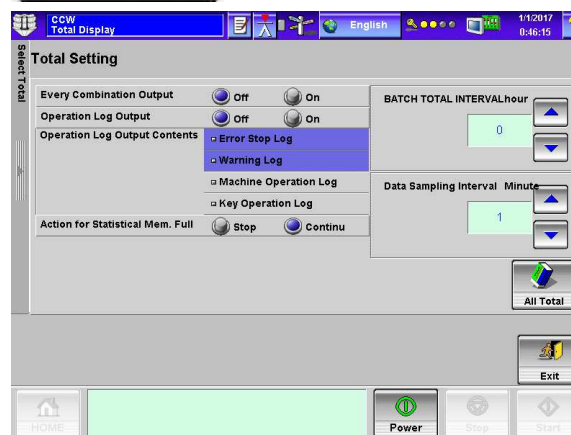
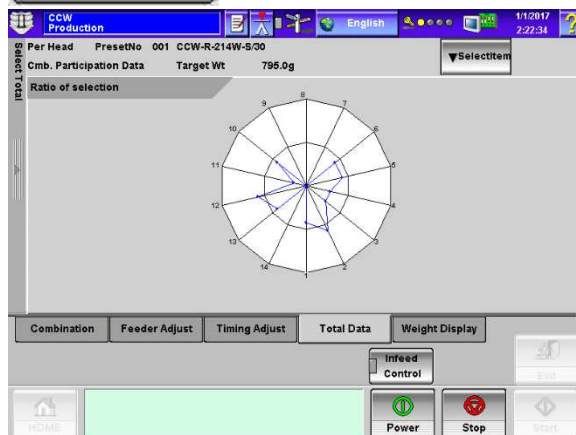
Weigher's Transitional Data

Per Head – Cmb.
Participation Data

Total Setting



Operation Log



Home Screen/Production Pop-Up Menu

~ Total Setting

Every Combination Output

*Will print every weight combination. Default is set to OFF.

Operation Log Output

*Will print each Operation Log Output Contents. Default is set to OFF.

NOTE: selection of the Operation Log Output Contents must be selected before the Operation Log Output will be printed.

Operation Log Contents

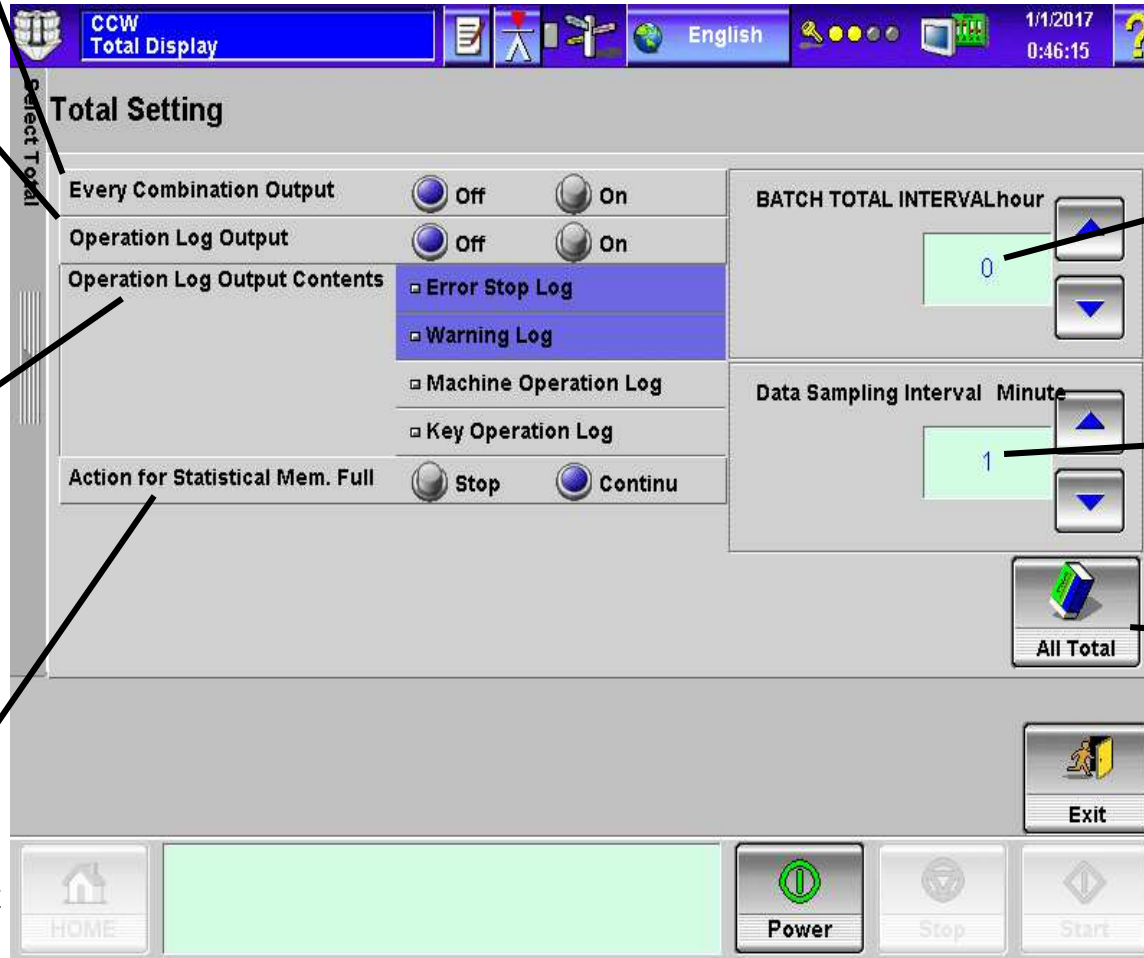
*Selects the various information that will be "logged" to the Operation Log. Default has no selections.

NOTE: the Operation Log will be able to log the first 100 selected logs, once reached, the oldest information will be written over.

Action for Statistical Memory Full

*The action that will be performed when the statistical information is at the maximum memory capacity. Default is set to STOP.

NOTE: if set to STOP and the memory reached maximum capacity, a Total Overflow Error will occur.



Batch Interval Selection (Hours)

Default is set to 0.

Data Sampling Interval (Minutes)

Default is set to 1.

Clear All Totals

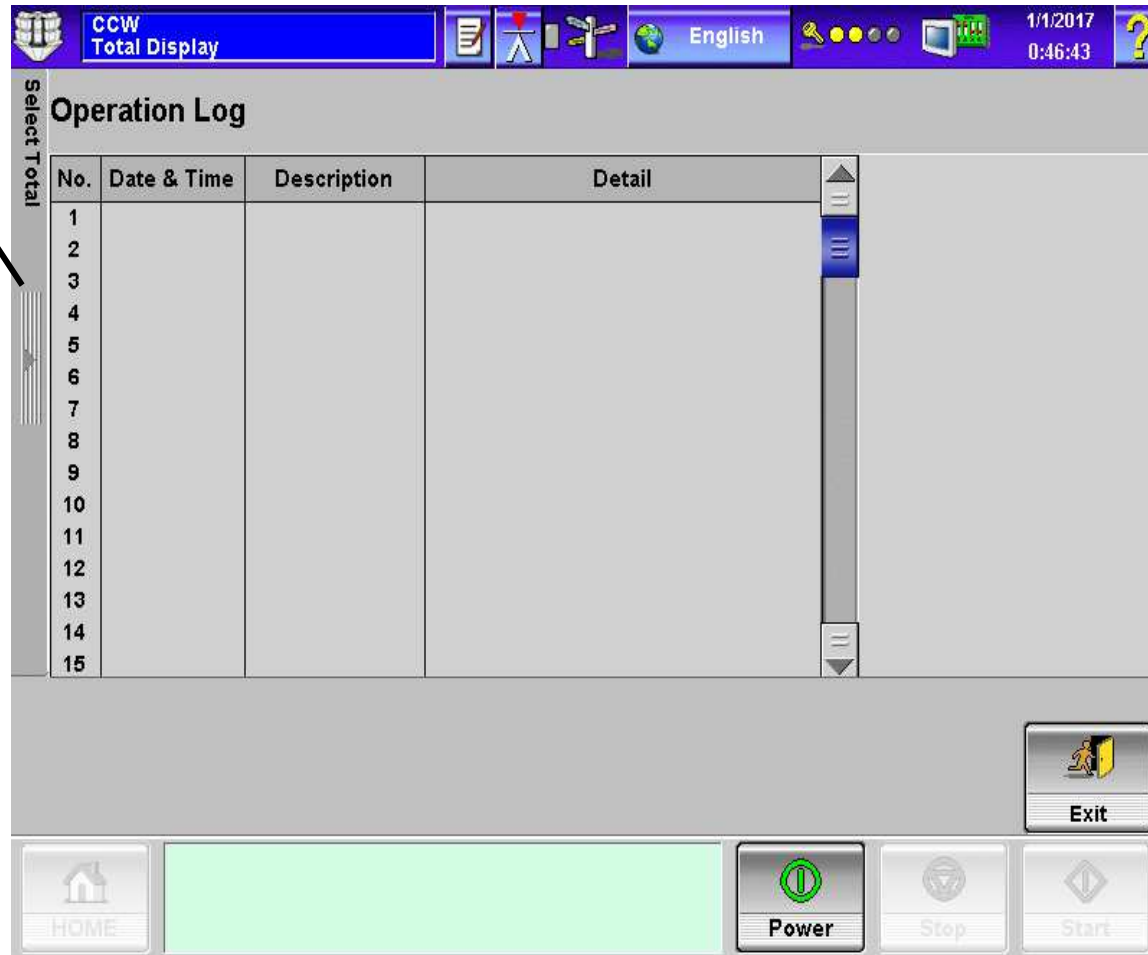
*will clear ALL statistical data from the Histogram. New data will start to be collected from the time in which the data was cleared. NOTE: once data is cleared, old data will no longer be available for review.

Home Screen/Production Pop-Up Menu

~ Operation Log

Display Viewing Area

*will log information that is selected from the Operation Log Output Contents (see page 37). Date & Time along with Description and Details will be presented in the Operation Log. A maximum of 100 logs will be viewable. Once the operation log is full, the oldest logged information will be written over with new log information.

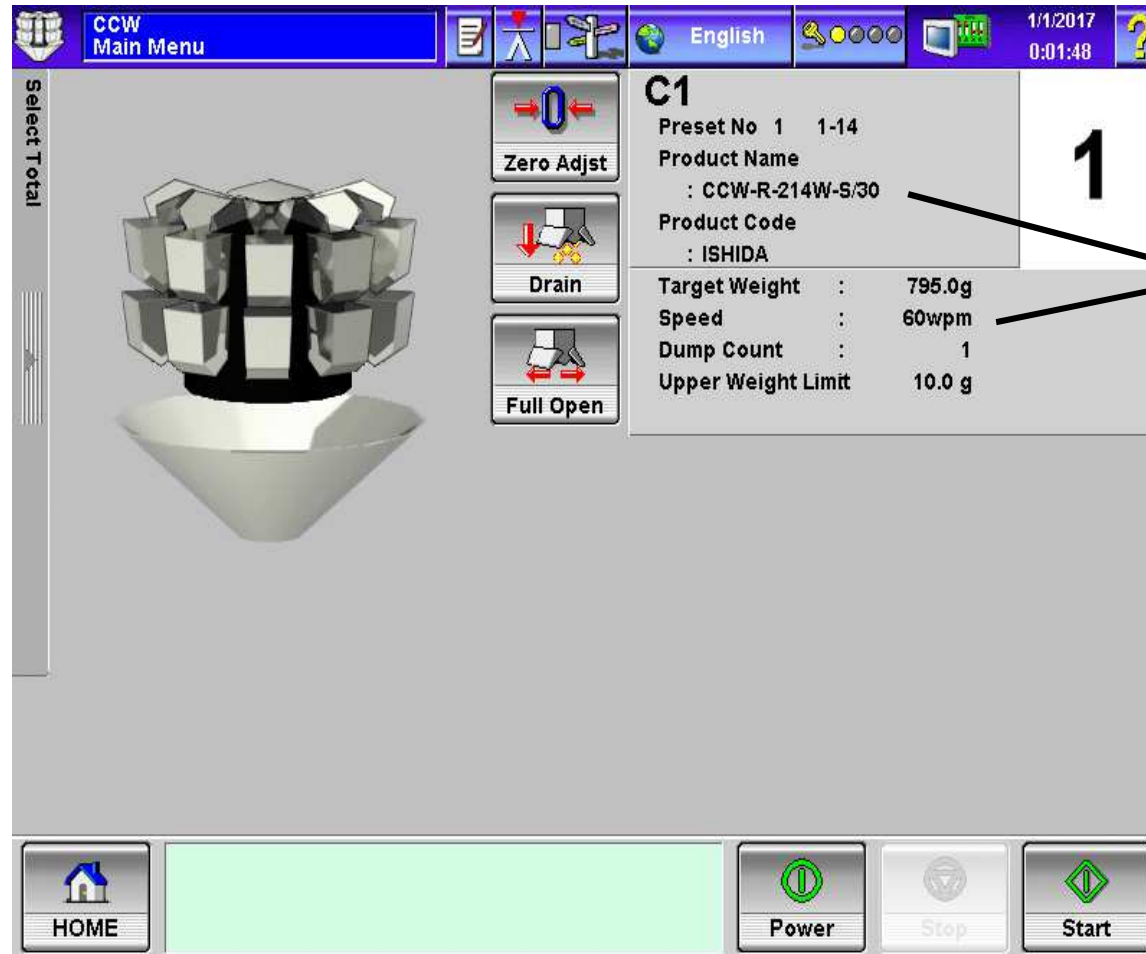


No.	Date & Time	Description	Detail
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			



Start Production

Start Production



*NOTE: Production Values can be altered or changed only when **BLUE**. NOTE: Must be in Site Engineer Level or higher to make a change to the values.

Production ~ Combination

Weigh Hopper Indicators

*Under normal operation:

1 Selected

*Selected the BEST combination of Weigh Hoppers closest to the Target Weight. NOTE: The weight of each individual WH must not exceed that of the Target Weight and the Upper Limit.

1 Available

*The weight in the WH is stable and is considered available to be selected for a combination calculation.

△ Auto Zero

*At the conclusion of the sixth combination, a selected WH will be automatically "re-zeroed." The weight in the WH, at the time of Auto Zero, must be +/- the weight value in the Auto Zero Tolerance.

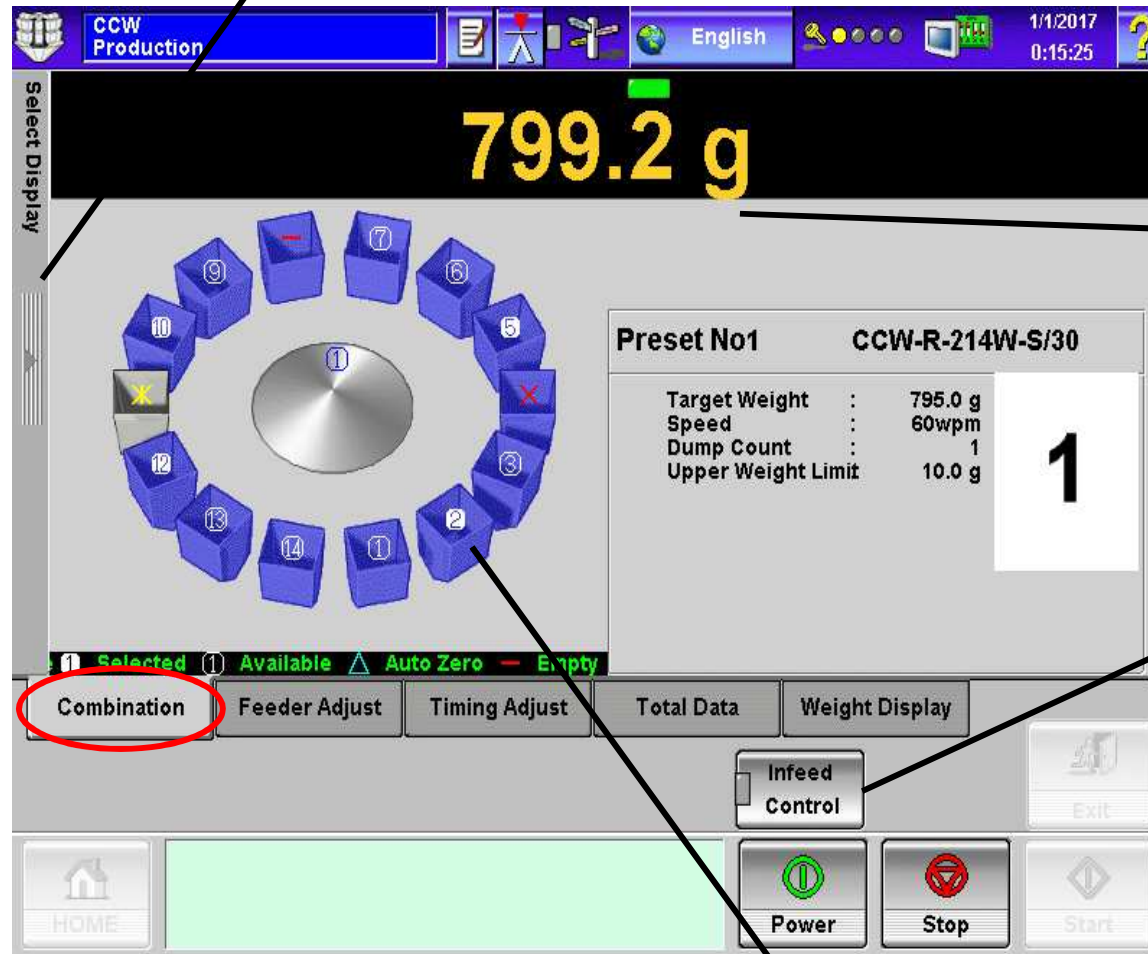
Unavailable

*The weight in the WH is not stable, as such, is not available for a combination calculation.

X Deactivated

*The WH has been turned OFF, as such, the WH is not available for combination calculation.

Select Display Key



Weight Display

The weight that has been discharged to the packaging system. NOTE: for a Proper discharge, the weight MUST be between the Target Weight and the Upper/Lower Weight Limits. For a combination that is ABOVE (Overweight) or BELOW (Underweight) a value of 0.0 g/oz. will be displayed.

Infeed Control Key

*Under normal operation leave the Infeed Control Key ON. However, at the conclusion of production, when the CCW is in Low Product, setting to OFF will allow the weigher to ignore DF Low Weight Percentage and production will resume.

Weigh Hopper Display

*The weight of the WH must be equal to or exceed the weight value in the Empty Judgement Weight..

*The WH that is in a state of error.

Production Keys and WH Display Symbols

No.	Name	Function
1	Combination weight display 	The combined weight value is displayed on each channel. The lamp indication varies depending on the results. Green lamp: Proper weight Yellow lamp: Over weight Red lamp: Under weight
2	Head condition display 	Displays the combination and weighing status.  : Head which was weighed and is stable.  : Head which was weighed, was selected for combination and discharged products. The color of the circle indicates the head stability, and there are three levels: green (most stable), white (more stable) and yellow (stable).  : Empty head. Blank: Unstable head.  : Head under the auto zero adjustment.  : Head with error of overload or jam.  : Head that is set to be deactivated.

Production

~ Radial Feeder Adjust

*Selects the TIME for the RF to vibrate. Will increase or decrease the “horizontal flow” as product moves along the Radial Feeder.

Amplitude *Selects the AMP for the RF to vibrate. Will increase or decrease the “vertical bounce” as product moves along the Radial Feeder.

Time

Increase Value

*Will increase the TIME or AMP values incrementally per one step.

Decrease Value

*Will decrease the TIME or AMP values incrementally per one step.

Select All Feeders

*Will select all Radial Feeders. **NOTE: under normal operation, when highlighted BLUE, selection is ON or is available to be adjusted. GRAY is normally OFF or not adjustable.**

Individual Feeder Select

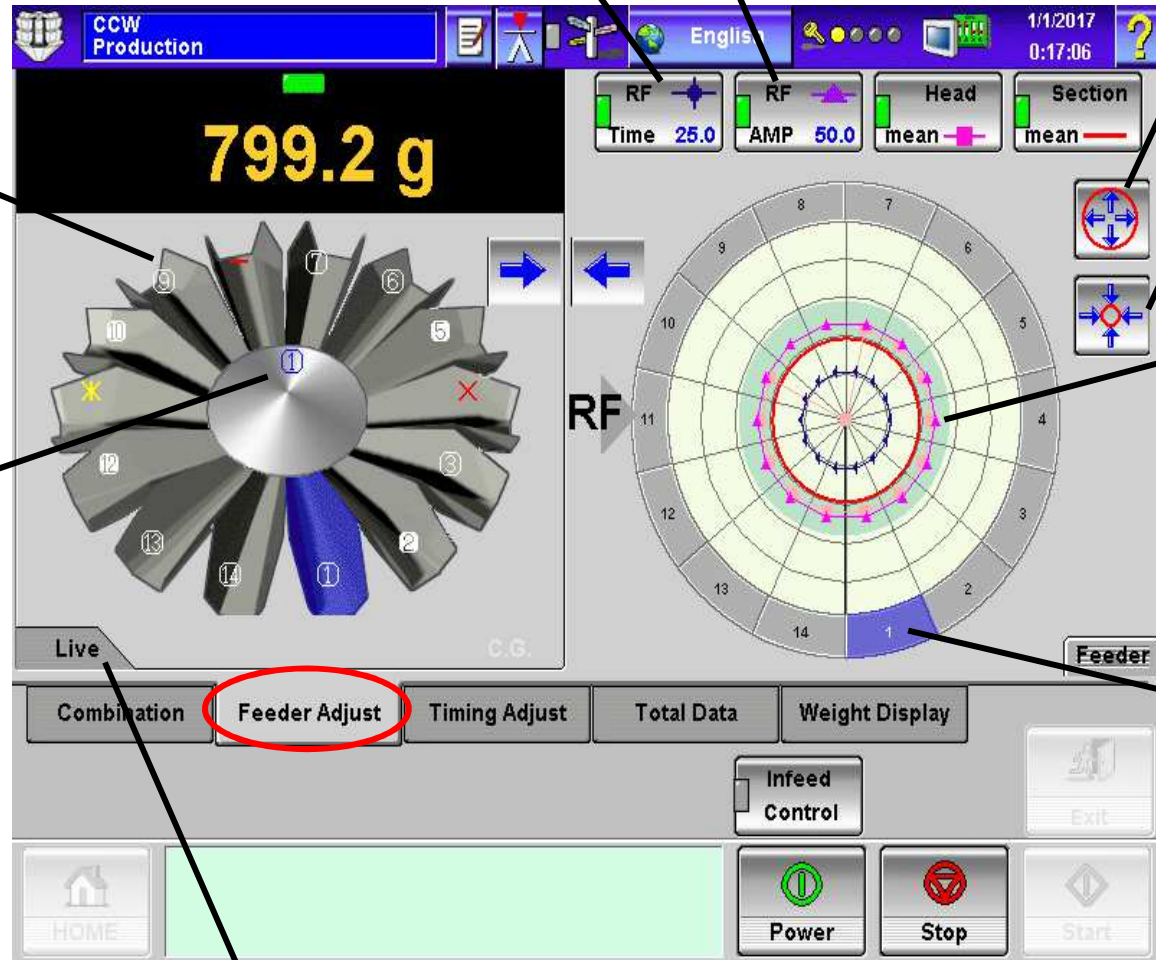
*Will select individual Radial Feeders. **NOTE: under normal operation, when highlighted BLUE, selection is ON or is available to be adjusted. GRAY is normally OFF or not adjustable.**

Individual Feeder Select

*Will select individual Radial Feeders. **NOTE: under normal operation, when highlighted BLUE, selection is ON or is available to be adjusted. GRAY is normally OFF or not adjustable.**

Dispersion Feeder Select

*Will select individual Dispersion Feeders. **NOTE: under normal operation, when highlighted BLUE, selection is ON or is available to be adjusted. GRAY is normally OFF or not adjustable.**



DF Camera On / Off

*Allows to toggle between the Dispersion Feeder Camera (Live) and the Computer Graphics (C.G.)

Production

~ Dispersion Feeder Adjust

*Selects the TIME for the DF to vibrate. Will increase or decrease the “horizontal flow” as product is dispersed across the Dispersion Feeder.

Time

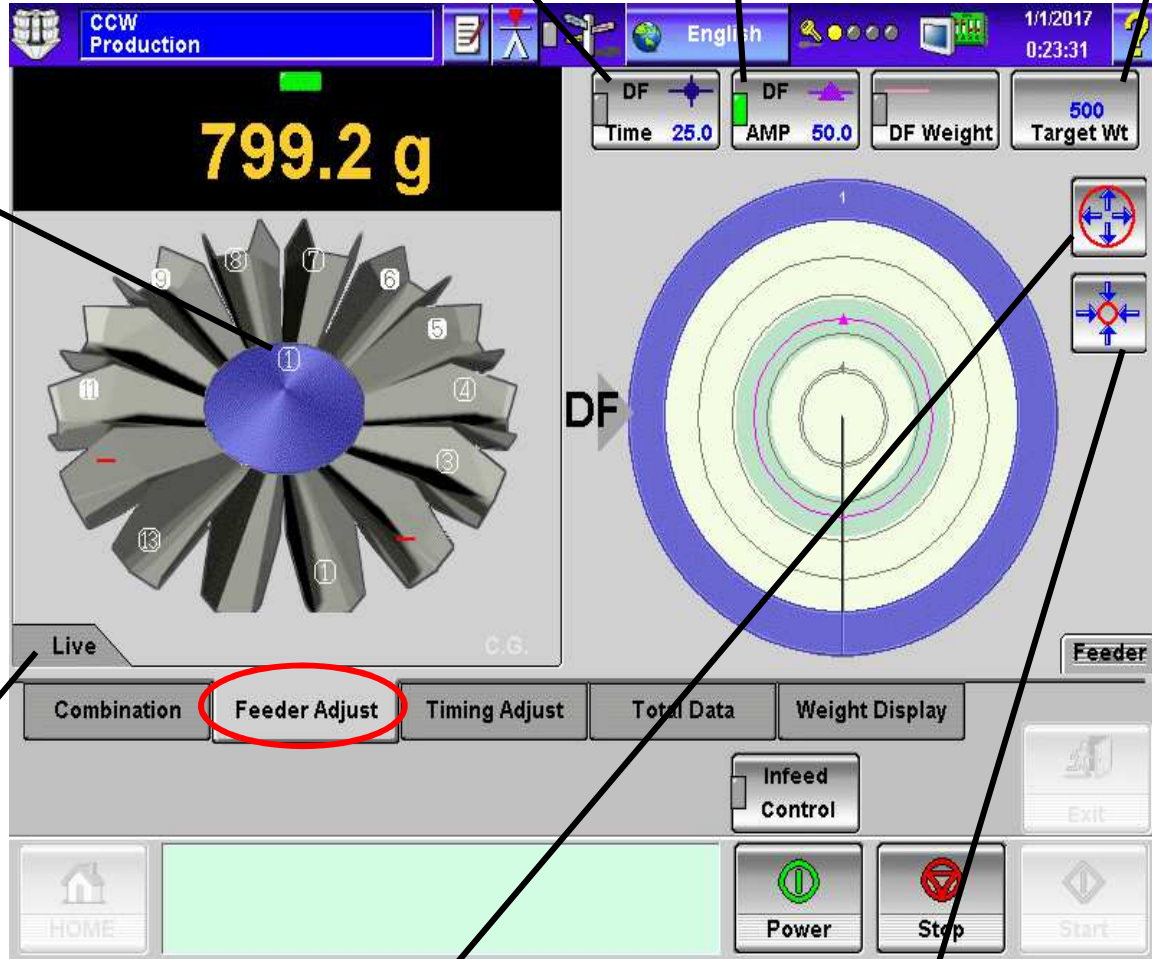
Amplitude

*Selects the AMP for the DF to vibrate. Will increase or decrease the “vertical bounce” as product is dispersed across the Dispersion Feeder.



Dispersion Feeder Select

*Will select individual Dispersion Feeders. **NOTE:** under normal operation, when highlighted BLUE, selection is ON or is available to be adjusted. GRAY is normally OFF or not adjustable.



DF Camera On / Off

*Allows to toggle between the Dispersion Feeder Camera (Live) and the Computer Graphics (C.G.)

A.

Maximum	5000
Minimum	0

DF Weight Setting: 500

Buttons: 7, 8, 9, CANCEL, 4, 5, 6, BS, 1, 2, 3, CLR, 0, -, +

*Will set the Dispersion Feeder Target Weight. The DF Load Cell will detect the weight on the DF and will “call” for the established DF weight setting.

B.

Maximum	100
Minimum	0

Upper Limit(%) Setting: 20

Buttons: 7, 8, 9, CANCEL, 4, 5, 6, BS, 1, 2, 3, CLR, 0, -, +

*Establishes the Upper Weight, in a percentage, from the DF Target Weight. Under normal operation, is 20%.

C.

Maximum	100
Minimum	0

Lower Limit(%) Setting: 20

Buttons: 7, 8, 9, CANCEL, 4, 5, 6, BS, 1, 2, 3, CLR, 0, -, +

*Establishes the Lower Weight, in a percentage, from the DF Target Weight. Under normal operation, is 20%.

Increase Value

*Will increase the TIME or AMP values incrementally per one step.

Decrease Value

*Will decrease the TIME or AMP values incrementally per one step.

Production

~ AFD Control Options



AFD (Automatic Feeder) Range Setting

*Sets the range in which the Upper and Lower Limit of the AMP and TIME can automatically be adjusted.
NOTE: The range can be set within values of 0~99.

Manual

*Adjustment of the AMP and TIME for the RF/DF will be controlled by the operator.

Efficiency Ref. Control

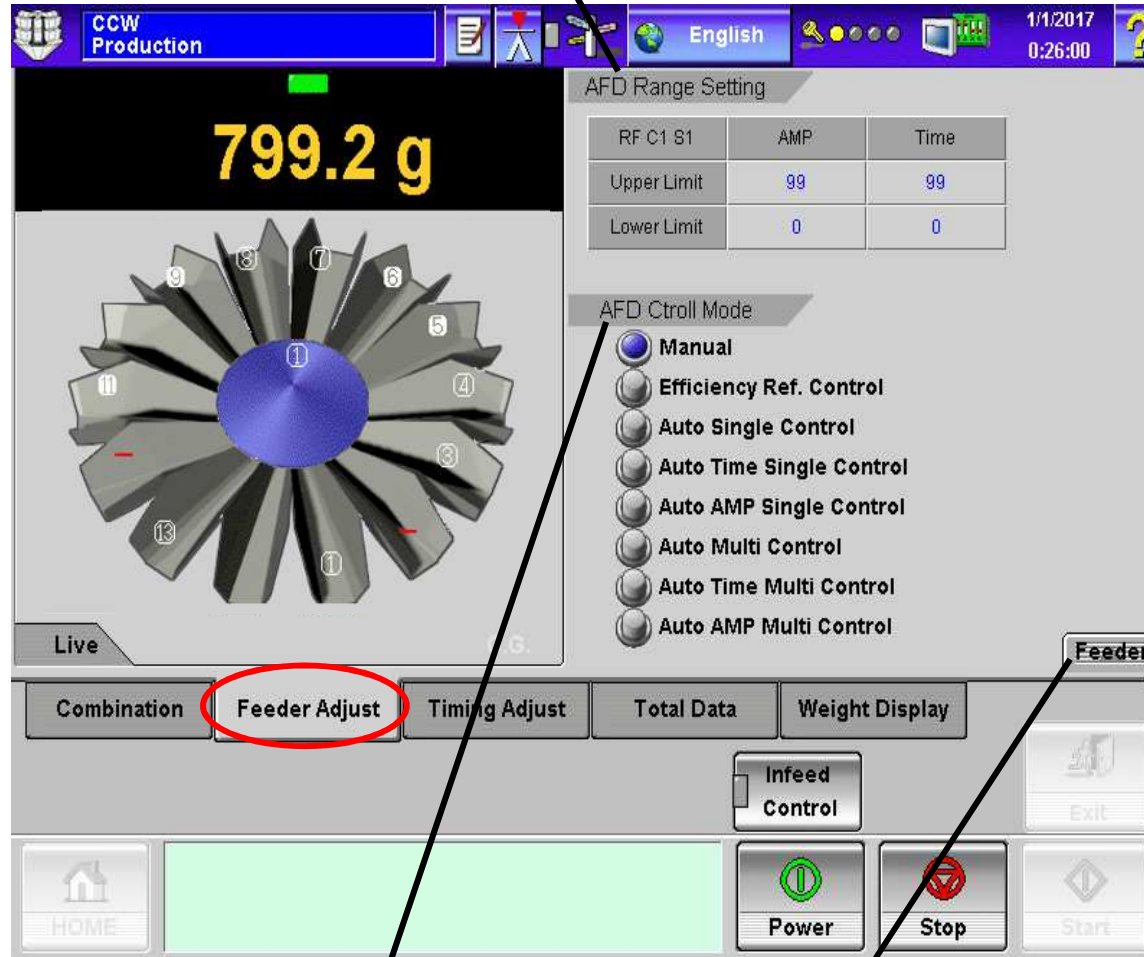
*Adjustment of the AMP and TIME for the RF/DF will be controlled by the operator, however, if the overall efficiency falls below the valued entered under the Good Efficiency Value (see Preset pg. 12) the AFD will assist with adjustments to the AMP and TIME of the RF/DF.

Auto Single Control

*Adjustment of the **AMP and TIME** for the RF/DF will be controlled by the AFD Control. The adjustments will be made to ALL RF/DF equally.

Auto Time Single Control

*Adjustment of the **TIME** for the RF/DF will be controlled by the AFD Control. The operator will control the AMP. The adjustments will be made to ALL RF/DF equally.



Auto AMP Single Control

*Adjustment of the **AMP** for the RF/DF will be controlled by the AFD Control. The operator will control the TIME. The adjustments will be made to ALL RF/DF equally.

Auto Multi Control

*Adjustment of the **AMP and TIME** for the RF/DF will be controlled by the AFD Control. The adjustments will be made to ALL RF/DF individually.

Auto Time Multi Control

*Adjustment of the **TIME** for the RF/DF will be controlled by the AFD Control. The operator will control the AMP. The adjustments will be made to ALL RF/DF individually.

Auto AMP Multi Control

*Adjustment of the **AMP** for the RF/DF will be controlled by the AFD Control. The operator will control the TIME. The adjustments will be made to ALL RF/DF individually.

AFD (Automatic Feeder)
Control Mode

Feeder Options Select

Production

~ Timing Adjust

HEAT AND CONTROL



Before making changes to values,
see page 47 for further explanation
on Timing Adjustments'

Auto Calculation

*Under normal operation, the Auto Calc. is set to OFF. When the Auto Cal. is set to OFF, no changes will be made to the various timing settings. However, if the Auto Cal. is set to ON when a change is made to the WH ON, the WH-PH will automatically be adjusted, and when a change is made to the PH ON, the PH-RF will automatically be adjusted.

NOTE: when changing the WH ON or the PH ON the automatic adjusted change will be equal to the time increase/decrease of the change made.

The screenshot displays the 'CCW Production' control interface. At the top, there's a status bar with 'English', '1/1/2017', and '0:19:12'. Below this is a table of timing settings:

Parameter	Value	Unit
IS-DS	0	ms
IS-TH1	0	
IS-DTH1	0	
IS-RS1	0	
RS ON	800	
IS-WH	0	
WH-PH	200	
PH-RF	60	
STAGGER	0	
WH ON	450	
PH ON	400	

Below the table, there's a 'Timing Adjust' button circled in red. To its right are buttons for 'Entr Time', 'Infeed Control', 'Power', 'Stop', and 'Start'. At the bottom, there's a 'HOME' button and a 'Auto Cal.' button with a slider set to 'On'.

Annotations on the interface include:

- A red starburst pointing to the top status bar with the text: "Before making changes to values, see page 47 for further explanation on Timing Adjustments'".
- A black arrow pointing from the 'Auto Calculation' section to the 'Timing Adjust' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Auto Cal.' button at the bottom.
- A black arrow pointing from the 'Timing Adjust' button to the 'Infeed Control' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Power' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Stop' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Start' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Entr Time' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Weight Display' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Total Data' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Feeder Adjust' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Combination' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Off Auto Cal.' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'On Auto Cal.' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Increase X 100ms' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Increase / Decrease X 10ms' button.
- A black arrow pointing from the 'Timing Adjust' button to the 'Decrease X 100ms' button.

Production

~ Timing Adjust

PH-RF – Sets the time delay from when the PH operation starts to when the RF operates to re-feed the PH.

WH-PH – Sets the time delay from when the WH operation starts to when the PH operates to re-feed the WH.

BH-WH – Sets the time delay from when the BH operation starts to when the WH operates to re-feed the BH

WH-BH – Sets the time delay from when the WH operation starts to when the BH operation start (used to “dump” product from both the WH and the BH).

Stagger – Sets the interval for discharging product from the selected heads used for that Combination Calculation.

(used when a timing hopper is not used)

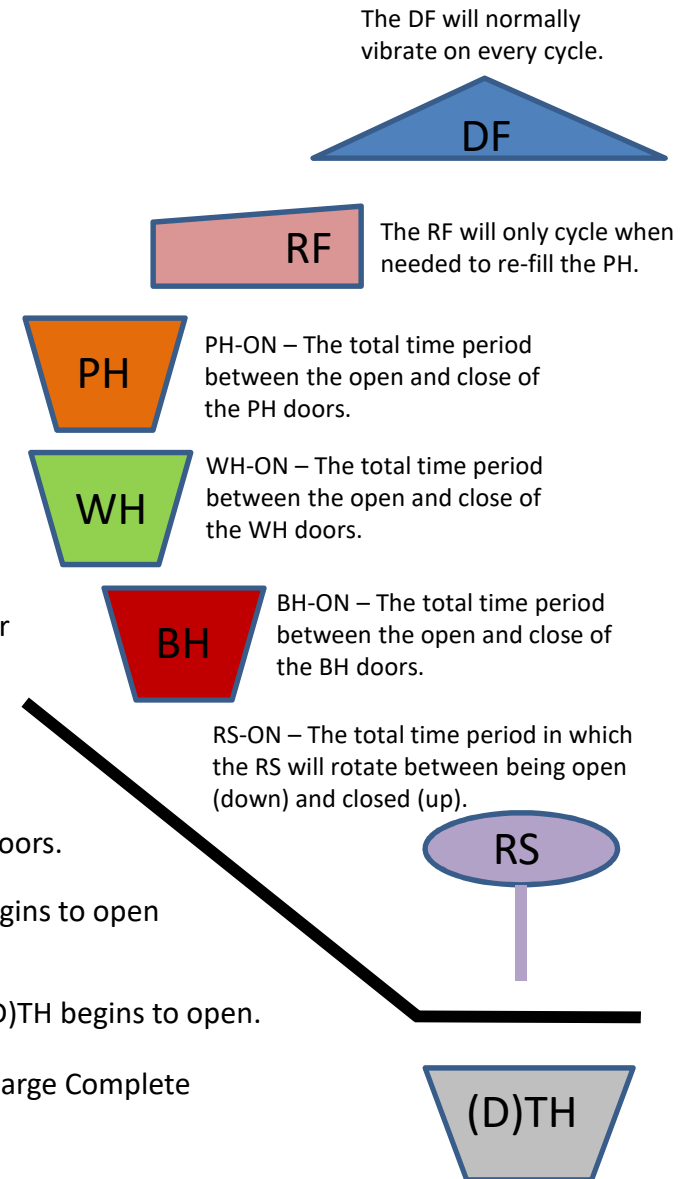
WH-DS – How long to wait to send the DS signal from when the WH door open.

IS-WH – How long to wait from getting the IS signal from the packer to open WH doors.

IS-RS – Sets the time delay from when the IS signal is received and when the RS begins to open (move in a downward direction).

IS-(D)TH – Sets the time delay from when the IS signal is received and when the (D)TH begins to open.

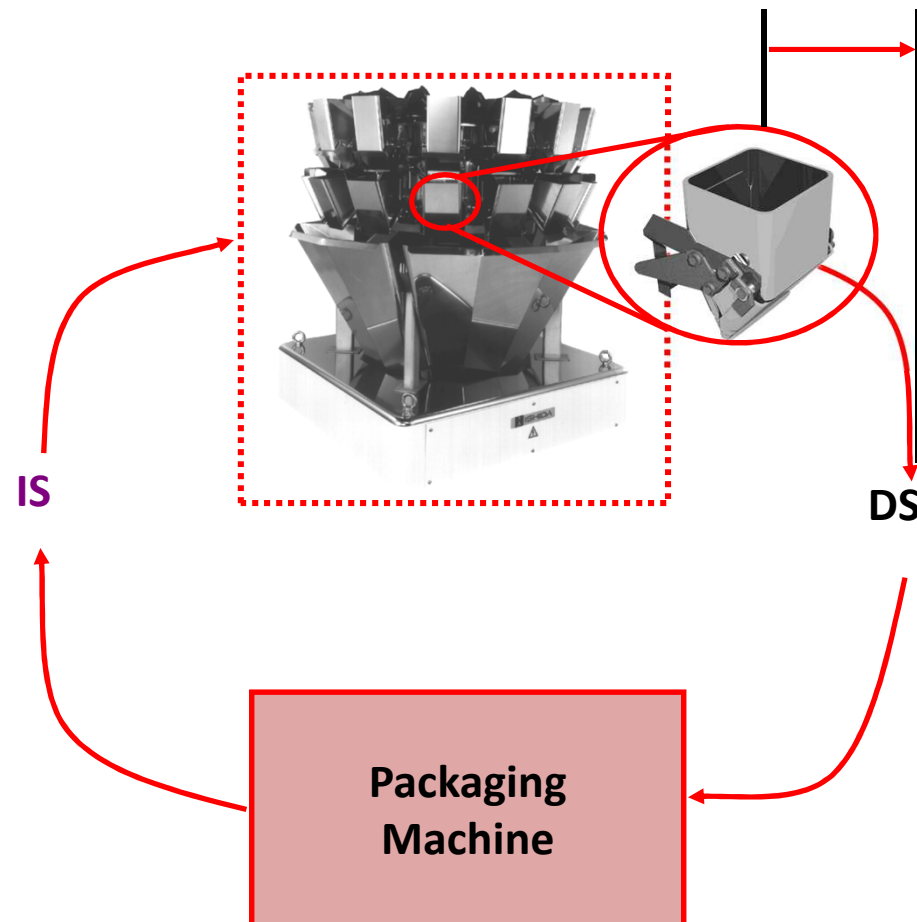
IS-DS – Sets the time delay from when the IS signal is received and when the Discharge Complete signal is sent.



TIMING ADJUST

WH - DS Weigh Hopper to Discharge complete Delay

(Product is getting “stuck” in the seal of the bag as it is making it to the bagger late)



WH - DS

=

↓ 'DROP TIME'

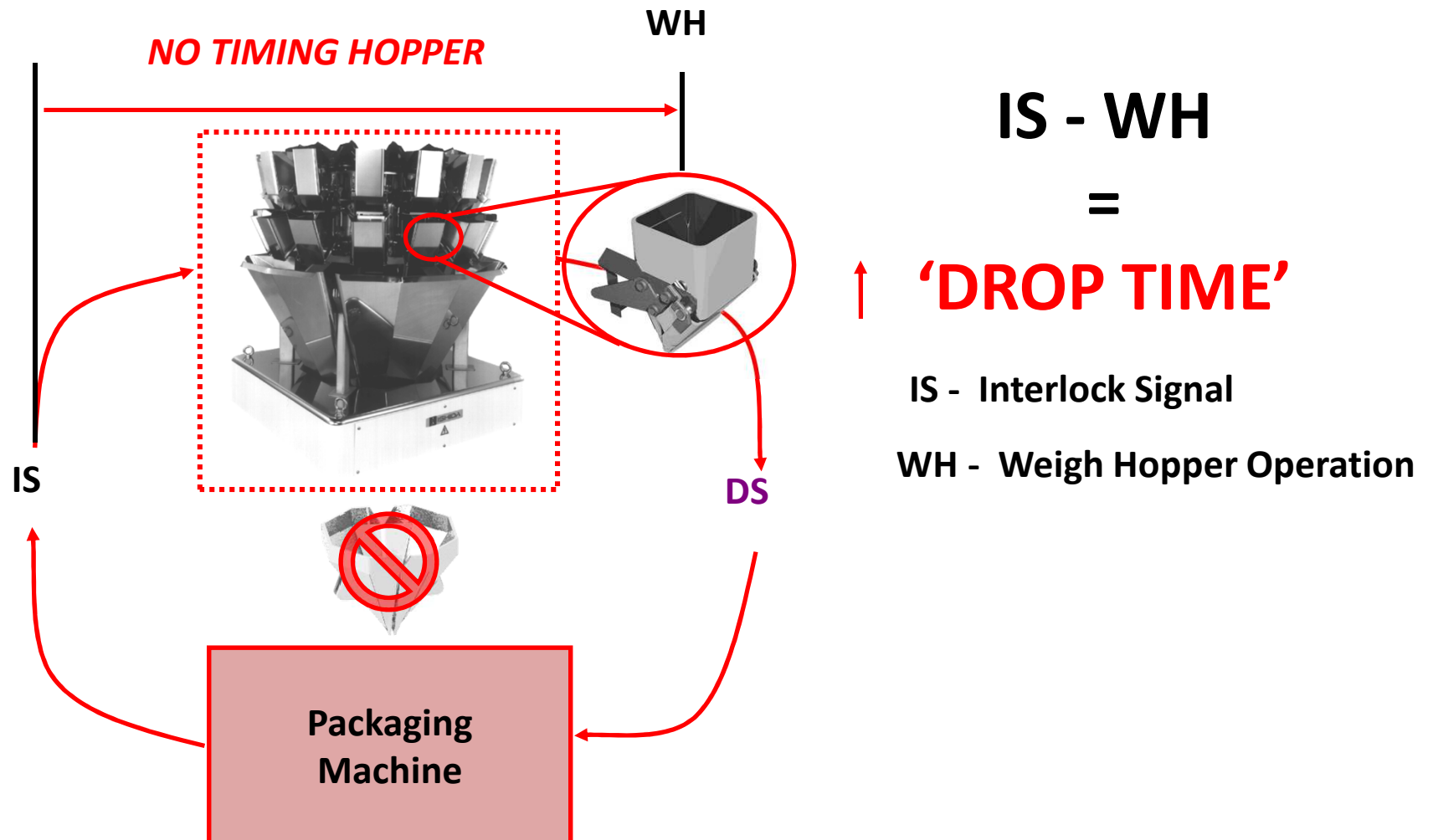
WH - Weigh Hopper Operation

DS - Discharge Complete Signal

TIMING ADJUST

IS - WH Interlock signal to Weigh Hopper Delay

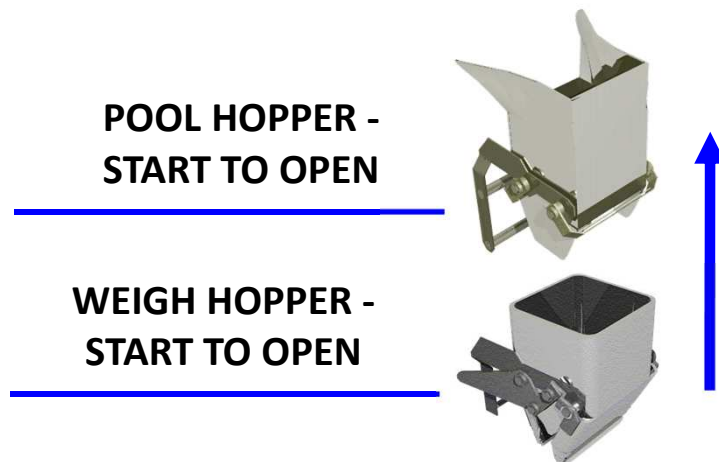
(Product is getting “stuck” in the seal of the bag as it is making it to the bagger early)



TIMING ADJUST

WH - PH Weigh Hopper to Pool Hopper Delay

REFILLING WEIGH HOPPER



WH - PH

Weigh Hopper Pool Hopper Delay

Delays the operation (start) of the Pool Hoppers after the Weigh Hoppers activates.

Should be set so that the Pool Hoppers operate after the Weigh Hopper

If product from the Pool hoppers are getting caught in the doors of the Weigh Hopper, increasing this delay may eventually "cure" the problem.

*RECOMMENDED TO NEVER BE SET TO ZERO;
as product may drop straight through WH*